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Electromagnetic Waves

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MCQ (Single Correct Answer)	39 questions
Answer key	at the end

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MCQ (Single Correct Answer) 39 questions

Q1. RE-NEET 2026

Open solution

The following table presents the part of the electromagnetic spectrum and their corresponding major applications.

	Part of the electromagnetic spectrum		Applications
P.	Microwave	I.	For purifying the water
Q.	UV rays	II.	For warming the food
R.	Gamma rays	III.	For AM and FM communication systems
S.	Radio wave	IV.	For treating the Cancer cells

The correct option is:

(A) P-II, Q-IV, R-III, S-I

(B) P-I, Q-II, R-III, S-IV

(C) P-I, Q-IV, R-II, S-III

(D) P-II, Q-I, R-IV, S-III

Q2. RE-NEET 2026

Open solution

An electromagnetic wave travelling in a lossless dielectric medium having a dielectric constant, $\varepsilon_r = 9$, has the electric field, $E_x = E_0 \sin\left(kz - 2\pi \times 10^6 t\right) \text{ V m}^{-1}$ where E_0 is the amplitude and k is the wave vector. Among the following options, the incorrect choice is

(A) The direction of propagation of the electromagnetic wave is along $+z$

(B) The speed of the electromagnetic wave inside the medium is 10^8 ms^{-1}

(C) The wavelength of the electromagnetic wave inside the medium is 300 m

(D) The magnetic field is given by the relation $B_y = \frac{B_0}{v}\sin\left(kz - 2\pi \times 10^6 t\right)$ where v is the speed of the electromagnetic wave inside the medium

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Q3. NEET 2026

Open solution

Match List I with List II:

List-I (Electromagnetic wave)		List-II (Production)	
A.	Microwave	I.	Electrons in atoms emit light when they move from a higher energy level to a lower energy level
B.	Visible light	II.	Radioactive decay of nucleus
C.	Gamma rays	III.	Vibration of atoms and molecules
D.	Infra-red rays	IV.	Klystron valve or magnetron valve

Choose the correct answer from the options given below:

(A) A-III, B-I, C-II, D-IV

(B) A-III, B-IV, C-I, D-II

(C) A-IV, B-I, C-II, D-III

(D) A-IV, B-III, C-II, D-I

Q4. NEET 2025

Open solution

The electric field in a plane electromagnetic wave is given by $E_z = 60\cos\left(5x + 1.5 \times 10^9 t\right)\mathrm{V/m}$ Then expression for the corresponding magnetic field is (here subscripts denote the direction of the field) :

(A) $Bz = 60\cos\left(5x + 1.5 \times 10^9 t\right)T$

(B) $B_y = 60\sin\left(5x + 1.5 \times 10^9 t\right)T$

(C) $B_y = 2 \times 10^{-7}\cos\left(5x + 1.5 \times 10^9 t\right)T$

(D) $B_x = 2 \times 10^{-7}\cos\left(5x + 1.5 \times 10^9 t\right)T$

Q5. NEET 2024 (Re-Examination)

Open solution

The electromagnetic radiation which has the smallest wavelength are

(A) X-rays

(B) Gamma rays

(C) Ultraviolet rays

(D) Microwaves

Q6. NEET 2024 (Re-Examination)

Open solution

If the ratio of relative permeability and relative permittivity of a uniform medium is $1:4$. The ratio of the magnitudes of electric field intensity (E) to the magnetic field intensity (H) of an EM wave propagating in that medium is (Given that $\sqrt{\frac{\mu_0}{\varepsilon_0}} = 120\pi$):

(A) $30\pi : 1$

(B) $1 : 120\pi$

(C) $60\pi : 1$

(D) $120\pi : 1$

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Q7. NEET 2024

The property which is not of an electromagnetic wave travelling in free space is that:

(A) They are transverse in nature

(B) The energy density in electric field is equal to energy density in magnetic field

(C) They travel with a speed equal to $\frac{1}{\sqrt{\mu_0 \varepsilon_0}}$

(D) They originate from charges moving with uniform speed

Open solution

Q8. NEET 2023

In a plane electromagnetic wave travelling in free space, the electric field component oscillates sinusoidally at a frequency of 2.0×10^{10} Hz and amplitude $48\ \mathrm{Vm}^{-1}$. Then the amplitude of oscillating magnetic field is : (Speed of light in free space $= 3 \times 10^8\ \mathrm{m\,s}^{-1}$)

(A) $1.6 \times 10^{-8}\ \mathrm{T}$ (B) $1.6 \times 10^{-7}\ \mathrm{T}$ (C) $1.6 \times 10^{-6}\ \mathrm{T}$ (D) $1.6 \times 10^{-9}\ \mathrm{T}$

Open solution

Q9. NEET 2022 Phase 2

The magnetic field of a plane electromagnetic wave is given by $\overrightarrow{B} = 3 \times 10^{-8} \cos(1.6 \times 10^3 x + 48 \times 10^{10} t)\hat{j}$, then the associated electric field will be :

(A) $9 \cos(1.6 \times 10^3 x + 48 \times 10^{10} t)\hat{k}$ V/m

(B) $3 \times 10^{-8} \cos(1.6 \times 10^3 x + 48 \times 10^{10} t)\hat{i}$ V/m

(C) $3 \times 10^{-8} \sin(1.6 \times 10^3 x + 48 \times 10^{10} t)\hat{i}$ V/m

(D) $9 \sin(1.6 \times 10^3 x - 48 \times 10^{10} t)\hat{k}$ V/m

Open solution

Q10. NEET 2022 Phase 2

The ratio of the magnitude of the magnetic field and electric field intensity of a plane electromagnetic wave in free space of permeability μ_0 and permittivity ε_0 is (Given that c - velocity of light in free space)

(A) $\frac{\sqrt{\mu_0 \varepsilon_0}}{c}$ (B) c

(C) $\frac{1}{c}$ (D) $\frac{c}{\sqrt{\mu_0 \varepsilon_0}}$

Open solution

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Q11. NEET 2022 Phase 1

Match List - I with List - II

	List - I (Electromagnetic waves)		List - II (Wavelength)
(a)	AM radio waves	(i)	10^{-10} m
(b)	Microwaves	(ii)	10^{2} m
(c)	Infrared radiations	(iii)	10^{-2} m
(d)	X-rays	(iv)	10^{-4} m

Open solution

Choose the correct answer from the options given below

(A) (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)

(B) (a) - (iii), (b) - (ii), (c) - (i), (d) - (iv)

(C) (a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)

(D) (a) - (ii), (b) - (iii), (c) - (iv), (d) - (i)

Q12. NEET 2022 Phase 1

When light propagates through a material medium of relative permittivity ε_r and relative permeability μ_r, the velocity of light, v is given by (c-velocity of light in vacuum)

Open solution

(A) v = c

(B) $v = \sqrt{\frac{\mu_r}{\varepsilon_r}}$

(C) $v = \sqrt{\frac{\varepsilon_r}{\mu_r}}$

(D) $v = \frac{c}{\sqrt{\varepsilon_r \mu_r}}$

Q13. NEET 2021

For a plane electromagnetic wave propagating in x-direction, which one of the following combination gives the correct possible directions for electric field (E) and magnetic field (B) respectively?

Open solution

(A) $-\hat{j}+\hat{k}, -\hat{j}+\hat{k}$

(B) $\hat{j}+\hat{k}, \hat{j}+\hat{k}$

(C) $-\hat{j}+\hat{k}, -\hat{j}-\hat{k}$

(D) $\hat{j}+\hat{k}, -\hat{j}-\hat{k}$

Q14. NEET 2020 Phase 1

The ratio of contributions made by the electric field and magnetic field components to the intensity of an electromagnetic wave is : (c = speed of electromagnetic waves)

Open solution

(A) 1 : 1 (B) 1 : c (C) 1 : c^2 (D) c : 1

Q15. NEET 2019

Which colour of the light has the longest wavelength?

Open solution

(A) blue (B) violet (C) red (D) green

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Q16. NEET 2018

An EM wave is propagating in a medium with a velocity $\vec{v} = v\hat{i}$. The instantaneous oscillating electric field of this em wave is along +y axis. Then the direction of oscillating magnetic field of the em wave will be along

(A) −z direction (B) +z direction (C) −y direction (D) −x direction

Open solution

Q17. NEET 2017

In an electromagnetic wave in free space that root mean square value of the electric field is E_{rms} = 6 V m^{-1}. The peak value of the magnetic field is

(A) 2.83×10^{-8} T (B) 0.70×10^{-8} T (C) 4.23×10^{-8} T (D) 1.41×10^{-8} T

Open solution

Q18. NEET 2016 Phase 1

Out of the following options which one can be used to produce a propagating electromagnetic wave ?

(A) A chargeless particle (B) An accelerating charge

(C) A charge moving at constant velocity (D) A stationary charge

Open solution

Q19. AIPMT 2015

The energy of the em waves is of the order of 15 keV. To which part of the spectrum does it belong ?

(A) Ultraviolet rays (B) γ-rays (C) X-rays (D) Infra-red rays

Open solution

Q20. AIPMT 2015 Cancelled Paper

A radiation of energy 'E' falls normally on a perfectly reflecting surface. The momentum transferred to the surface is (C = Velocity of light)

(A) $\frac{2E}{C^2}$ (B) $\frac{E}{C^2}$ (C) $\frac{E}{C}$ (D) $\frac{2E}{C}$

Open solution

Q21. AIPMT 2014

Light with an energy flux of 25×10^4 W m^{-2} falls on a perfectly reflecting surface at normal incidence. If the surface area is 15 cm^2, the average force exerted on the surface is

(A) 1.25×10^{-6} N (B) 2.50×10^{-6} N (C) 1.20×10^{-6} N (D) 3.0×10^{-6} N

Open solution

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Q22. NEET 2013 (Karnataka)

An electromagnetic wave of frequency $\upsilon = 3.0$ MHz passes from vaccum into a dielectric medium with relative permittivity ε_r = 4.0. Then

(A) Wavelength is doubled and frequency becomes half.

(B) Wavelength is halved and frequency remains unchanged.

(C) Wavelength and frequency both remain unchanged.

(D) Wavelength is doubled and frequency unchanged.

Open solution

Q23. NEET 2013

The condition under which a microwave oven heats up a food item containing water molecules most efficiently is

(A) Microwaves are heat waves, so always produce heating.

(B) Infra-red waves produce heating in a microwave oven.

(C) The frequency of the microwaves must match the resonant frequency of the water molecules.

(D) The frequency of the microwaves has no relation with natural frequency of water molecules.

Open solution

Q24. AIPMT 2012 Mains

The ratio of amplitude of magnetic field to the amplitude of electric field for an electromagnetic wave propagatting in vacuum is equal to

(A) the speed of light in vacuum

(B) reciprocal of speed of light in vacuum

(C) the ratio of magnetic permeability to the electric susceptibility of vacuum

(D) unity

Open solution

Q25. AIPMT 2012 Prelims

The electric field associated with an em wave in vacuum is given by

E = $\hat{i}$ 40 cos (kz $-$ 6 $\times$ 10^8 t) where E, z and t are in volt/m, meter and seconds respectively. The value of wave vector k is

(A) 2 m^{-1} (B) 0.5 m^{-1} (C) 6 m^{-1} (D) 3 m^{-1}

Open solution

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Q26. AIPMT 2011 Prelims

The electric and the magnetic field, associated with an e.m. wave, propagating along the +z-axis, can be represented by

(A) $\left[\overrightarrow{E} = E_0\hat{i}, \overrightarrow{B} = B_0\hat{j}\right]$

(B) $\left[\overrightarrow{E} = E_0\hat{k}, \overrightarrow{B} = B_0\hat{i}\right]$

(C) $\left[\overrightarrow{E} = E_0\hat{j}, \overrightarrow{B} = B_0\hat{i}\right]$

(D) $\left[\overrightarrow{E} = E_0\hat{j}, \overrightarrow{B} = B_0\hat{k}\right]$

Open solution

Q27. AIPMT 2011 Prelims

The decreasing order of wavelength of infrared, microwave, ultraviolet and gamma rays is

(A) microwave, infrared, ultraviolet, gamma rays

(B) gamma rays, ultraviolet, infraref, microwaves

(C) microwaves, gamma rays, infrared, ultraviolet

(D) infrared, microwave, ultraviolet, gamma rays

Open solution

Q28. AIPMT 2010 Mains

The electric field of an electromagnetic wave in free space is given by $\overrightarrow{E} = 10\cos(10^7 t + kx)\hat{j}\ V/m$, where t and x are in seconds and metres respectively. It can be inferred that

(1) the wavelength λ is 188.4 m.

(2) the wave number k is 0.33 rad/m.

(3) the wave amplitude is 10 V/m.

(4) the wave is propagating along +x direction.

Which one of the following pairs of statements is correct ?

(A) (3) and (4) (B) (1) and (2) (C) (2) and (3) (D) (1) and (3)

Open solution

Q29. AIPMT 2010 Prelims

Which of the following statements is false for the properties of electromagnetic waves ?

(A) Both electric and magnetic field vectors attain the maxima and minima at the same place and same time.

(B) The energy in electromagnetic wave is divided equally between electric and magnetic vectors.

(C) Both electric and magnetic field vectors are parallel to each other and perpendicular to the direction of propagation of wave.

(D) These waves do not require any material medium for propagation.

Open solution

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Q30. AIPMT 2009

The electric field part of an electromagnetic wave in a medium is represented by $E_x = 0$;

$E_y = 2.5\frac{N}{C}\cos\left[\left(2\pi \times 10^6 \frac{rad}{s}\right)t - \left(\pi \times 10^{-2}\frac{rad}{m}\right)x\right]$;

$E_z = 0$. The wave is

(A) moving along x direction with frequency 10^6 Hz and wavelength 100 m.

(B) moving along x direction with frequency 10^6 Hz and wavelength 200 m.

(C) moving along $-$x direction with frequency 10^6 Hz and wavelength 200 m.

(D) moving along y direction with frequency $2\pi \times 10^6\ Hz$ and wavelength 200 m.

Open solution

Q31. AIPMT 2008

The velocity of electromagnetic radiation in a medium of permittivity ε_0 and permeability μ_0 is given by

(A) $\frac{1}{\sqrt{\mu_0\varepsilon_0}}$

(B) $\sqrt{\frac{\mu_0}{\varepsilon_0}}$

(C) $\sqrt{\frac{\varepsilon_0}{\mu_0}}$

(D) $\sqrt{\mu_0\varepsilon_0}$

Open solution

Q32. AIPMT 2007

The electric and magnetic field of an electromagnetic wave are

(A) in opposite phase and perpendicular to each other

(B) in opposite phase and parallel to each other

(C) in phase and perpendicular to each other

(D) in phase and parallel to each other.

Open solution

Q33. AIPMT 2005

If λ_v, λ_x and λ_m represent the wavelengths of visible light, X-rays and microwaves respectively, then

(A) $\lambda_m > \lambda_x > \lambda_v$

(B) $\lambda_m > \lambda_v > \lambda_x$

(C) $\lambda_v > \lambda_x > \lambda_m$

(D) $\lambda_v > \lambda_m > \lambda_x$

Open solution

Q34. AIPMT 2003

We consider the radiation emitted by the human body. Which one of the following statements is true?

(A) The radiation emitted is in the infra-red region

(B) The radiation emitted only during the day

(C) The radiation is emitted during the summers and absorbed during the winters

(D) The radiation emitted lies in the ultraviolet region and hence is not visible

Open solution

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Q35. AIPMT 2003

Which of the following rays are not electromagnetic waves ?

(A) X-rays (B) γ-rays (C) β-rays (D) heat rays

Open solution

Q36. AIPMT 2002

The velocity of electromagnetic wave is parallel to

(A) $\vec{B} \times \vec{E}$ (B) $\vec{E} \times \vec{B}$

(C) $\vec{E}$ (D) $\vec{B}$

Open solution

Q37. AIPMT 2002

What is the cause of Green house effect ?

(A) infra-red rays (B) ultra violet rays

(C) X-rays (D) radio waves.

Open solution

Q38. AIPMT 2001

Biological importance of ozone layer is

(A) it stops ultraviolet rays

(B) ozone layer reduces green house effect

(C) ozone layer reflects radio waves

(D) ozone layer controls O_2/H_2 ratio in atmosphere.

Open solution

Q39. AIPMT 2000

The frequency order for γ-rays (a), X-rays (b), UV rays (c) is

(A) $b > a > c$ (B) $a > b > c$ (C) $c > b > a$ (D) $a > c > b$

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Answer Key

1. **D**	2. **C**	3. **C**	4. **C**	5. **B**
6. **C**	7. **D**	8. **B**	9. **A**	10. **C**
11. **D**	12. **D**	13. **C**	14. **A**	15. **C**
16. **B**	17. **A**	18. **B**	19. **C**	20. **D**
21. **B**	22. **B**	23. **C**	24. **B**	25. **A**
26. **A**	27. **A**	28. **D**	29. **C**	30. **B**
31. **A**	32. **C**	33. **B**	34. **A**	35. **C**
36. **B**	37. **A**	38. **A**	39. **B**	

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